THE STUDENTS INNOVATION POWER LAB

" TECHNICAL STANDARDS"
Common product Specification

MASTER PAGE

Welcome to Student Innovation Lab. The custom student-innovation, skill-hub architectures, and custom combo assemblies contained in this catalog are built upon unified design framework utilizing a standardized Common Controlling Unit, structurally engineered to meet or exceed internal technical and performance baselines.

MASTER PRODUCTION CATALOG

SL-PB-01 | Ultra-Slim 18650 Nano Power Bank with Original Li-ion 3200-3500 mAh

2 c DMECG /3c PANASONIC –Single Cell Version

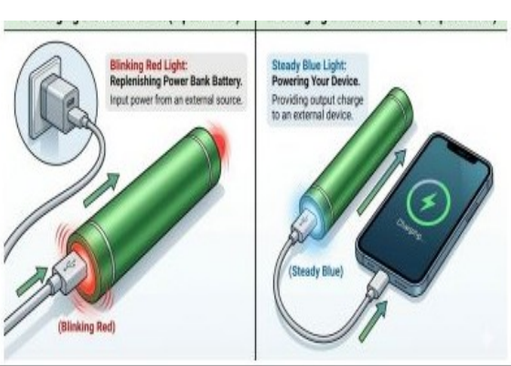
Sleek Aluminum Nano Capsule Design - ***Smallest Power Bank***

(Powerful with Stylish Look)

DESIGNED AND ASSEMBLED FOR EDUCATION PURPOSE ONLY
(NOT FOR SALE)

Cell Core Optimization: Exclusively engineered and optimized for genuine, high-density Panasonic or DMECG Lithium cells (32E Model). Housed inside a premium, ultra-compact cylindrical capsule aluminum body where sleek aesthetics meet next-generation as per Common Product Specification.

 Capsule Form Factor: This highly efficient, pocket-friendly capsule design functions exactly as a heavy-duty, standard power bank but with a significantly minimized footprint.





MADE IN INDIA
STUDENT INNOVATION POWER

CLICK TO VIEW - DATA SHEET PRODUCT VIDEO

Images displayed in this catalog are for visual representation only. Exact technical metrics and context-specific parameters should be obtained from the Data Sheets.

ASSEMBLED BY THE STUDENTS INNOVATION POWER LAB –
"Where Next-Gen Engineering Meets Passion"
MADE IN INDIA

DESIGNED AND ASSEMBLED FOR EDUCATION PURPOSE ONLY
(NOT FOR SALE)

SL-EP-05 | Executive Professional Planner Matrix

The Ultimate ON GRID / GRIDLESS Traveler Portfolio

Portfolio Integration: An elegant Executive Professional Planner portfolio that seamlessly unfolds to reveal a comprehensive array of SL-PB-01 **Ultra-Slim 18650 Nano Power Bank** utilizing original high-density DMECG / Panasonic Lithium cells, delivering fast-charging designed to meet the Next Gen Engineering parameters as per Common Product Specification.



Accompanied rechargeable cable array - Detachable Solar Matrix (optional**),**

optimized on AC charging with grid less connectivity even when totally isolated from the electrical utility grid.

CLICK TO VIEW DATA SHEET PRODUCT VEDIO

Images displayed in this catalog are for visual representation only. Exact technical metrics and context-specific parameters should be obtained from the Data Sheets.

ASSEMBLED BY THE STUDENTS INNOVATION POWER LAB –
"Where Next-Gen Engineering Meets Passion"
MADE IN INDIA

SL-CC-04 | Power Plus Pro Universal Power Bank
***All-In-One* Solution**
with 30A Controller *Management Matrix*

DESIGNED AND ASSEMBLED FOR EDUCATION PURPOSE ONLY
(NOT FOR SALE)

System Capacity: Heavy-duty, robust 30A capacity energy management system managed by Smart Logic Controller for highly accurate, error-free operations as per Common Product Specification Details.

		
	The All in One Power Bank comes with Ultra-Slim 18650 Nano Power Bank with Li-ion (original) DMECG Li-ion Power Pack (14.8V -16.8V) Pro Range NMC N18650 Continuous discharge Rating 2C Configuration 4S1P Sustain up to continuous load 65 W	
Images displayed in this catalog are for visual representation only. Exact technical metrics and context-specific parameters should be obtained from the Data Sheets		
CLICK TO VIEW	DATA SHEET	PRODUCT VEDIO

.

ASSEMBLED BY THE STUDENTS INNOVATION POWER LAB –
"Where Next-Gen Engineering Meets Passion"
MADE IN INDIA

SECTION 2

FLOOD LIGHT INVERTER LIGHT - LED EXTENDER
Power Cuts Gone, Light Always On

FOR EDUCATION PURPOSE ONLY
(NOT FOR SALE)

SL-FL-02 - Flood Light (30W to 200W) SL – CL-06- **Ceiling Light (**30W to 00W)

Waterproof IP65 with Smart Logic and remote-controlled sensor

Unlimited backup: Magical Backup (Lithium Iron Phosphate LiFePO4) is **rechargeable** up to 3000-5000 lifecycle and is **replaceable** for re use again with zero electricity costs and **Dual Charging Option** of balanced AC Grid Charging option with Gridless connectivity.

Detachable Solar Panel with Cable —**Detachable cable** allows the Lithium Iron Phosphate (**LiFePO4)** Battery to be placed inside / under shadow away while, SOLAR PANEL is kept under open Sun Light to harvests massive renewable energy.

This works as inverter Light for whole night with unlimited backup for Home / Garden / Remote Area / Sports Arena (for both indoor / outdoor as Heavy duty Flood Light.





-

ASSEMBLED WITH VERIFIED COMPONENTS ONLY BY STUDENT INNOVATION POWER PROJECT
"Where Next-Gen Engineering Meets Passion"
MADE IN INDIA

SL-LE-07 LED EXTENDER - *Power Cuts Gone, Light Always On*

DESIGNED AND ASSEMBLED FOR EDUCATION PURPOSE ONLY
(NOT FOR SALE)

LED EXTENDER – Magical device to maximize Brightness with back up time of any home-bulbs /Emergency Bulbs
This is a Magical Device allows to run standard home-bulbs /Emergency Bulbs with same Brightness with Unlimited ***backup time with Zero Electricity Cost.***
Why other Emergency Bulb turns to DIM Mode during Power Cut and LED Extender sustain same brightness during power cut

The Emergency Bulb has only by a Tiny single low volt inbuilt battery inside of lower Amp an can store a nominal back up while charged during power on and automatically goes to Dim Candlelight Mode during power cut.

The System Logic of LED Extender is run by a full fledged powerful controller to constantly drws Power from Lithium Iron Phosphate (LiFePO4) ***with*** Magical Backup from (12.8 V -14.6 V) 6000 mAh 6A - 3C Discharge rated (18A) with Extreme Power Preservation /Efficiency (90%). It is **rechargeable** up to 3000-5000 lifecycle and is **replaceable** for re use again with an additional feature of **Zero electricity costs** and **Dual Charging Option** from Griddles **Solar Panel with alternatives** of balanced AC Grid Charging Via Adaptor with

Regulated DC Socket Set Up A – Supports any standard DC home-bulbs / Emergency Bulbs up to Maximum Load capacity up to 230 W, Run Time 79 Wh

AC Socket Set Up B - The AC Socket (220 V 50 Herts) supports to run any standard AC home-bulbs / AC Emergency Bulbs (like Philips, Jaguar etc.) or comfortably support other local /generic AC Bulbs up to Maxim Load 100 W Run Time 55 Wh with utmost thermal protection..

LED Extender Kit comes with (All inclusive) – Compact design of one small Controlling Unit with Lithium Iron Phosphate-LifePO4 Pack, Solar Panel, Charging Adopter (AC to DC Balanced)

Provided with the Kit – 2 Packs (total 10 Pcs) 12 V (DC) Hanging Bulb comprised 5 Pcs 12 W DC LED Bulbs and 5 Pcs 9 W LED Bulbs compatible with the DC Socket only.

SL – DC-08- DC LED Bulbs - DC 12 V 12 W 50000 Hrs Max 1300 Lumens brightness 1300 Lumens in transformation of vision friendly Cdm2.

ADVICE - Use either AC Bulb with AC Socket distinguished as SET UP A (AC) or use DC Bulb with DC Socket SET UP B (DC)

CAUTION : Do not try to plug in 12 V DC Bulb into AC Socket (220 V), the internal circuitry balance of bulb will damage / fry immediately.

SECTION 3
INVERTER BLDC FAN

DESIGNED AND ASSEMBLED FOR EDUCATION PURPOSE ONLY
(NOT FOR SALE)

SL-SF-09 PORTABLE FAN WITH **SOLAR**
Solar Panel
Compact light wight Battery pack
Rechargeable Portable Fan

For best use in vehicle / small size room.

SL-SF-10 Inverter Table Fan with Solar (unlimited back up with zero electric cost)

UPCOMING SOLAR INVERTER BLDC CEILING FAN

Best use for Electricity cost saving and as Emergency /Inverter during Power Cut

SL-SF-11 - Inverter BLDC Solar Inverter Table Fan with LED Extender Combo

For those who can not afford the cost of Heavy Inverter / Hybrid Solar System

SECTION 4

MINI POWER HUB (Hybrid Grid Solar Appliance
with Dual Charging Options (AC Grid support)

SL-PH-12: The ideal, budget-friendly power solution for those who want reliable backup without the high cost or footprint of a heavy-duty inverter system. The Mini Power Hub is an all-in-one portable power station designed to seamlessly run standard home LED bulbs, emergency lights, BLDC ceiling/table fans, laptops, TVs, and mobile devices.

COMMON SYSTEM PLATFORM PARAMETERS

(Standardized Infrastructure for both AC/DC Set up)

Compact Controlling Unit: Heavy-duty, industrial-grade protection chassis featuring an integrated **30A Universal Automated Intelligent Controller** with maximum thermal performance headroom.

Energy Storage Engine: Premium **12.8V 30Ah (384 Wh Nominal) Lithium Iron Phosphate (LiFePO)** cell matrix.. Chemically supports **instantaneous Power Threshold** up to **768 Watts** of raw cell delivery.

Discharge Rate Capability: Heavy-duty continuous **2C scaling capacity (60 Amps)** supporting appliance loads.

Photovoltaic Generator: Premium, high-efficiency **100W Mono PERC / Flexible Solar Panel** manufactured by elite Tier-1 manufacturer **Waaree Energies**.

Solar Recharge Benchmark: 5 Hours of direct, standard Indian daylight (7:00 AM – 5:00 PM) for a complete 0 to 100% chemical replenishment loop.

Dual-Charging Matrix: Equips an alternative, balanced **220V AC Grid-to-DC Smart Adapter** for automated, safe charging from utility lines during rainy seasons or low-sunlight intervals.

Ultra-Efficiency DC Core Line:

Engineered specifically for efficient off-grid load management., the system maintains a 90% ultra-efficient net path rating, converting a full 345.6 Wh into usable power.

Output Configuration: Regulated Plug-and-Play DC Setup (A)
12V / 9V DC Ports: Dedicated terminal block for high-illumination home DC LED Bulbs, Flood Lights, and DC-wound BLDC ceiling or table fans.

5V USB Arrays: Dual high-speed, dedicated ports for smart phones and mobile communication accessories.

19V Laptop Terminal: Dedicated **5.5mm DC Jack** running laptops (45W–65W) via an automated Plug-and-Play Universal DC-to-DC Boost Adapter.
12V TV Port: Heavy-gauge barrel connection configured to drive high-efficiency **32-inch Native DC LED TVs (30W–40W)** directly from the battery bus.

Mainstream 220V AC Household Integration

Engineered for seamless integration with standard residential wiring and consumer household products, utilizing a premium **Pure Sine Wave Inverter Sub-Module** (Compatible with Microtek Super Power, Luminous Eco Watt Neo 700, or Exide Inverterz lines).

Hardware Upper Ceiling Cap: Strictly governed at a continuous **500W Maximum AC Load** threshold by the inverter's diagnostic microchip to ensure total electronic protection.

Inversion Efficiency Footprint: Operating at a standard **75% net inverter path efficiency**, yielding **288 Wh** of usable AC energy

Output Configuration: Standard AC Socket Setup (B)
220V 50Hz Dual Sockets: Driven via a heavy-duty **700VA transformer bus** to supply clean, silent, identical-to-grid power directly to legacy AC home appliances.

Run time by Solar Power only (an alternative, balanced **220V AC Grid-to-DC Smart Adapter** is provided for extended use or safe charging during rainy day /extreme low-sunlight / cloudy intervals for both DC Set up (supports DC Appliances) and AC Line with inverter (optional) :

Load	Run time during Day		Night time (including Power Cut)	
	DC Set up for DC Appliances	AC Line with inverter optional	DC Set up for DC Appliances	AC Line with inverter optional
2 LED Bulbs 2 BLDC Fan	Whole day	Whole day	5.3 Hours	3.2 Hrs
2 LED Bulbs 2 BLDC Fan, Laptop / 32-inch LED TV	8.6 Hours	4.2 Hours	3.1 Hours	2.1 Hrs

System The Power Station Module Product Module includes (after un boxing) 384 Wh LiFePO4 Box + 100 W Waree Panel + Grid Adopter compatible with DC Set up only and AC Set with Inverter is is optional and sold separately as per price chart.

The TECHNICAL STANDARDS , Durability, Back up Battery Life Cycle, Chassis Engineering and Product Specification and all are detailed in the MASTER PAGE (attached)

SECTION 5

CRAFTED WITH ELECTRONIC & DECORATIVE

DESIGNED AND ASSEMBLED FOR EDUCATION PURPOSE ONLY
(NOT FOR SALE)

Ultra-Low Current/ Zero Heat : All decorative assemblies are designed / (step-down /buck) to ultra-low current thresholds up to maximum (5V). This ensures zero heat buildup inside enclosed artistic crafts.

Weatherproof Sealing Protocols: All decorative craft frameworks are strictly integrated UV-stabilized, IP65-rated polymer sealants or clear silicone encapsulation over exposed circuit traces to eliminate short circuits during high-humidity atmospheric deployment.

Solar Arrays: Built to interface seamlessly with the Micro Solar Bag matrix, optimizing power draw via automated dusk-to-dawn photovoltaic sensor switches.

==========

CRATED WITH ELECTRONICS (HAND MADE) & INTERIOR DECORATION

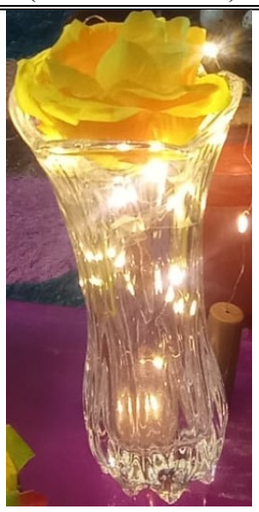


PRODUCT VIDEO DATA SHEET

SL –EC-14 Decorative Electro craft on Ceramic - Acrylic



PRODUCT VIDEO DATA SHEET



PRODUCT VIDEO DATA SHEET

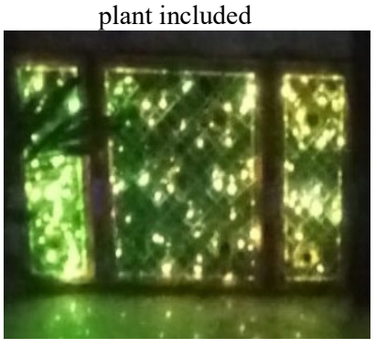
SL –EC-13 Decorative Electro craft - Flower Glass Vase





PRODUCT VIDEO DATA SHEET

SL –EC-15 Decorative Electro craft - Round Glass Vase with plant included

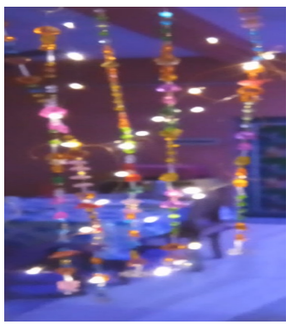


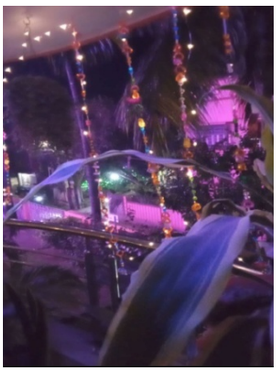
PRODUCT VIDEO DATA SHEET

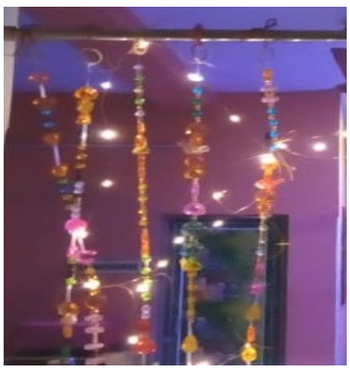
SL –EC-16 Decorative Electro Craft - Christmas (original Plant Pine) Tree

SL –EC-17 FIRE FLIES (movement and coloured are designed with Micro Electronically.

UPCOMING – FIRE FLIES Electronically.Movement with Sleeping Music Sound with rain







PRODUCT VIDEO DATA SHEET

SL –EC-18 Decorative Electro crafted Curtain - for Indoor / Balcony / Openterrace

MADE IN INDIA | ASSEMBLED BY THE STUDENT INNOVATION POWER PROJECT "Where Next-Gen Engineering Meets Passion"

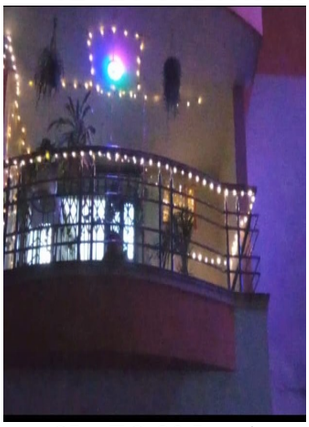






PRODUCT VIDEO DATA SHEET

SL –EC-19 Decorative Electro crafted Indoor Plant Pot (on ceramics / glass/ acrolyte)



SL –EC-20 Interior Decoraton for Balcony / Stair/ Open Terrace



UPCOMING SOON

BIRDS FOUNTAIN



SL –EC-21 Flower Light for outdoor garden

MADE IN INDIA | ASSEMBLED BY THE STUDENT INNOVATION POWER PROJECT
"Where Next-Gen Engineering Meets Passion"